



Smart Fertilizer Spraying System Using Plant Disease Severity Analysis

ABSTRACT

The increasing demand for precision agriculture requires smart systems for disease detection and optimized chemical usage. This paper presents a Smart Fertilizer Spraying System using deep learning to classify tomato disease severity (low, medium, high). A Raspberry Pi with a USB camera and YOLOv8 (Py Torch) model analyzes fruit images, while an Arduino controls fertilizer spraying time via relay pumps based on severity. The robotic platform enables automation. The system achieved 75–80% accuracy under controlled conditions and reduces fertilizer waste compared to uniform spraying, supporting efficient and sustainable agriculture.

EXISTING SYSTEM METHODS

- **Time-consuming process** when monitoring large agricultural fields.
- **Excessive use of fertilizers or pesticides**, which increases costs and environmental pollution.
- **Delayed disease detection**, which may lead to severe crop damage.

DISADVANTAGES OF EXISTING SYSTEM

- Over Use
- Inefficiency
- Pollution
- Imprecise
- Manual

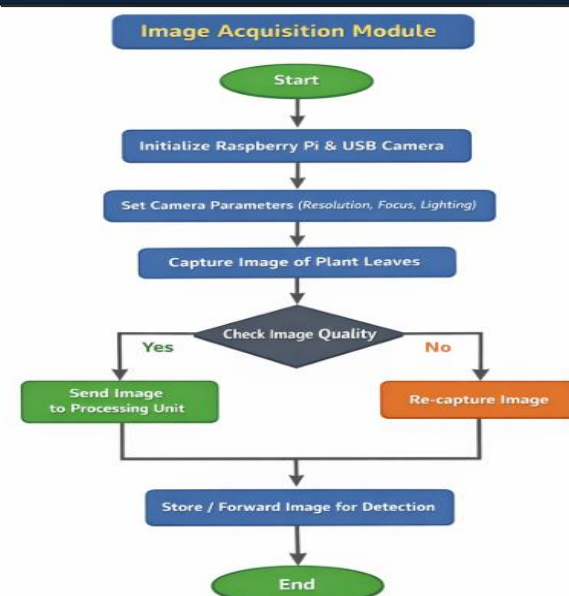
PROPOSED SYSTEM

This project proposes a Smart Fertilizer Spraying System using YOLOv8 for plant disease detection and severity analysis. A Raspberry Pi with a camera processes images, and based on severity (low, medium, high), an Arduino controls spraying duration via pumps and relays. The robot enables automated movement, powered by solar-supported battery and monitored via Bluetooth. The system reduces fertilizer waste, labor, and supports sustainable agriculture.

ADVANTAGES OF PROPOSED SYSTEM

1. Reduces Human Labor and Intervention
2. Minimizes Fertilizer Wastage
3. Enables Real-Time Monitoring and Decision Making.

FLOW CHART



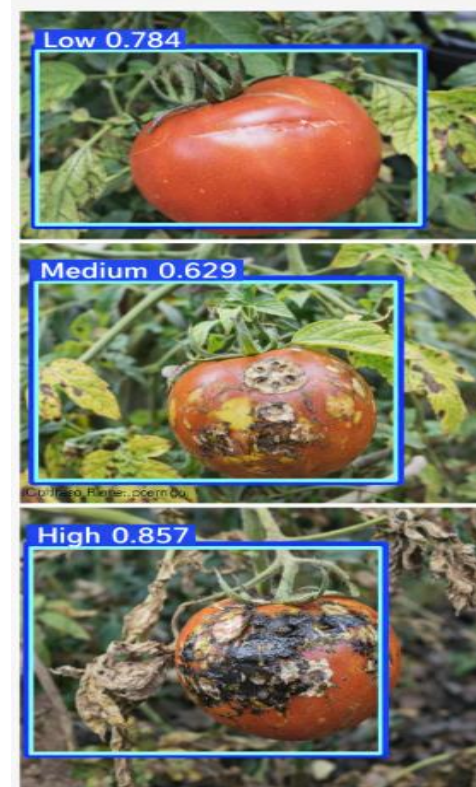
MODULES

- Image acquisition
- Disease Detection Module
- Control & Spraying Module
- Navigation & Communication Module

MODULES DESCRIPTION

1. **Image acquisition:** This module captures real-time images of plant leaves using a USB camera connected to the Raspberry Pi. The camera continuously monitors crops.
2. **Disease Detection Module:** Uses Raspberry Pi, USB camera, and YOLOv8 to capture and analyze plant images for detecting diseases and classifying severity levels.
3. **Control & Spraying Module:** Arduino controls pumps, relays, and nozzles to adjust fertilizer spraying based on detected severity (low, medium, high).
4. **Navigation & Communication Module:** Manages robot movement.

OUTPUT



CONCLUSION

The Smart Fertilizer Spraying System with Plant Disease Severity Analysis is an innovative solution for precision agriculture. It integrates image-based disease detection with automated fertilizer spraying to ensure accurate and need-based application, reducing wastage and improving crop health. The Raspberry Pi efficiently handles image processing, while the Arduino microcontroller manages hardware control, enabling smooth robot movement and reliable performance. Overall, the system minimizes human intervention, optimizes resource usage, and promotes eco-friendly farming, thereby improving agricultural productivity.

BATCH NO: B – 10

UNDER THE GUIDANCE OF

Dr. C. Sasikala

M. Tech., Ph.D

TEAM MEMBERS

B REVATHI 224G1A3380

G.SREEJA 224G1A33A9

Y.SRAVANI 224G1A33A4

Y.TRINESH REDDY 234G5A3311